Drainage and imbibition capillary pressure curves

Water-wet, oil-wet and mixed-wet rock: the equations and how they shape the curves

How the capillary pressure curve of a rock changes with wettability. Each curve is built from the two equations in Section 1, applied to a bundle of pores of different sizes. The curves are schematic, not measured data.

1. Capillary pressure

1.1 Pressure form

$$P_c = P_o - P_w \tag{1}$$

The pressure jump across the curved oil-water interface. Positive P_c means oil pressure is higher than water pressure; negative means the reverse. Only the difference matters: P_c rises if P_o rises or P_w falls.

1.2 Contact-angle form

$$P_c = \frac{2\sigma\cos\theta}{r} \tag{2}$$

In a cylindrical pore the interface is a spherical cap of radius R = r / cos θ, and Young–Laplace gives $P_c = 2\sigma / R$. σ is the oil-water interfacial tension (30 mN/m here), θ the contact angle measured through the water, and r the pore-throat radius (1 to 20 µm here).

1.3 Putting the two together

$$P_o - P_w = \frac{2\sigma\cos\theta}{r} \tag{3}$$

The left side is the applied pressure difference. The right side is the threshold of one pore, its capillary entry pressure.

- **Sign:** P_c has the sign of cos θ: positive for water-wet rock (θ < 90°), negative for oil-wet rock (θ > 90°).
- **Size:** $|P_c|$ is proportional to $1/r$, so small pores need a larger pressure difference.
- **Direction:** θ depends on the direction of motion (larger when water advances), which opens the hysteresis loop (Section 2.3).

1.4 Why θ is 0–90° for water-wet rock and 90–180° for oil-wet rock

$$\sigma_{os} - \sigma_{ws} = \sigma_{ow}\cos\theta \tag{4}$$

The subscripts are os = oil-solid, ws = water-solid, ow = oil-water. Since $\sigma_{ow} > 0$, cos θ has the sign of $\sigma_{os} - \sigma_{ws}$. **Water-wet:** the water-solid tension is lower, water spreads on the surface, the angle through the water is small, cos θ > 0. **Oil-wet:** the opposite, θ > 90°, cos θ < 0. At 90° the rock has no preference. Measured through the oil instead, the ranges swap ($\theta_{oil} = 180° - \theta$).

Example. σ = 30 mN/m, r = 5 µm, θ = 30°: $P_c = 2 \times 0.030 \times \cos 30° / (5 \times 10^{-6}) = 10.4$ kPa (1.5 psi). With θ = 150° it is −10.4 kPa.

Rock	θ (through water)	cos θ and P_c	Wetting phase
Water-wet	below 75°	positive	water
Intermediate	75° to 105°	near zero	neither
Oil-wet	above 105°	negative	oil
Mixed-wet	small pores water-wet, large pores oil-wet	changes sign with pore size	both, in different pores

2. Drainage and imbibition

Named by water saturation S_w (the usual reservoir convention). **Drainage:** S_w falls as oil enters water-filled pores. **Imbibition:** S_w rises as water enters oil-filled pores.

2.1 Each pore is a gate with a threshold

Oil enters a water-filled pore when

$$P_c \geq \frac{2\sigma\cos\theta_r}{r} \tag{5}$$

with θ_r the receding contact angle of water. Water enters an oil-filled pore when

$$P_c \leq \frac{2\sigma\cos\theta_a}{r} \tag{6}$$

with θ_a the advancing contact angle. Sweeping P_c opens the pores one by one; S_w is the water left in the pores not yet opened.

Rule for every wettability: drainage means P_c rises (S_w falls); imbibition means P_c falls (S_w rises). Wettability only shifts the whole loop up (water-wet, $P_c > 0$) or down (oil-wet, $P_c < 0$).

2.2 Pressure path: spontaneous and forced behaviour

The sign of the threshold in Equations 5 and 6 decides which processes the rock performs on its own and which must be driven by an imposed pressure difference. A positive threshold means the rock resists the invading fluid, so it must be pushed in. A negative threshold means capillary suction draws the fluid in without an external push.

Water-wet rock ($0° < \theta < 90°$, $\cos\theta > 0$). The loop lies in the positive P_c region.

- **Drainage.** Oil must displace the wetting phase, so P_o is raised above P_w. P_c increases as progressively smaller pores are entered.
- **Imbibition.** Water, the wetting phase, is drawn in spontaneously as P_w rises toward P_o. P_c decreases and approaches zero, where spontaneous imbibition ends. Raising P_w above P_o ($P_c < 0$) is possible, but this forced tail adds little saturation because the remaining oil is trapped.

Oil-wet rock ($90° < \theta < 180°$, $\cos\theta < 0$). The loop lies in the negative P_c region, and the rock is initially held water-saturated by $P_w > P_o$.

- **Drainage.** Oil, the wetting phase, enters spontaneously as P_o rises (or P_w falls) relative to the other. P_c increases toward zero while remaining negative, and no pressure beyond that is needed because the rock already favours oil. A forced tail ($P_c > 0$) can lower S_i slightly.
- **Imbibition.** Water, the non-wetting phase, must be forced in. P_w is raised (or P_o lowered), so P_c becomes more negative and returns close to its starting value, slightly below it because $\theta_a > \theta_r$.

Wettability indices (Amott-Harvey, USBM) compare the spontaneous and forced parts of a measured loop.

2.3 Why the two curves do not coincide

- **Contact-angle hysteresis.** $\theta_a > \theta_r$, so $\cos\theta_a < \cos\theta_r$ (for water-wet rock both are positive; only the size differs). Imbibition thresholds are lower than drainage thresholds, so the imbibition curve lies below. Here $\cos 50° / \cos 20° \approx 0.68$, a 32% gap, which is the whole gap in these figures. With $\theta_a = \theta_r$ and no trapping the curves would overlap.
- **Pore geometry (not modelled).** Drainage is controlled by narrow throats, imbibition by wider pore bodies and snap-off, so the effective radius differs between the processes.
- **Trapping.** Water snaps off and leaves oil blobs disconnected, so imbibition stops at $S_m = 1 - S_{or}$. Drainage turns vertical at S_i, the water held in films and the tiniest pores.

2.4 How the curves were built

Each rock is a bundle of 700 pores with a log-normal size distribution (1 to 20 µm, median 4 µm). Each pore gets a drainage threshold from Equation 5 and an imbibition threshold from Equation 6; S_w is the volume of pores not yet invaded, scaled between S_i and the end point. S_i and S_{or} are assumed. In the mixed-wet case the large pores (about 37% of pore volume) are oil-wet.

Rock	θ_r (drainage)	θ_a (imbibition)	S_i	S_{or}	S_m
Water-wet	20°	50°	0.20	0.20	0.80
Oil-wet	140°	160°	0.10	0.30	0.70
Mixed-wet	20° (r < 5 µm) 130° (r ≥ 5 µm)	50° (r < 5 µm) 160° (r ≥ 5 µm)	0.15	0.10	0.90

3. Water-wet rock

$\theta_r = 20°$, $\theta_a = 50°$: $\cos\theta > 0$ in every pore, so P_c is positive for the whole loop.

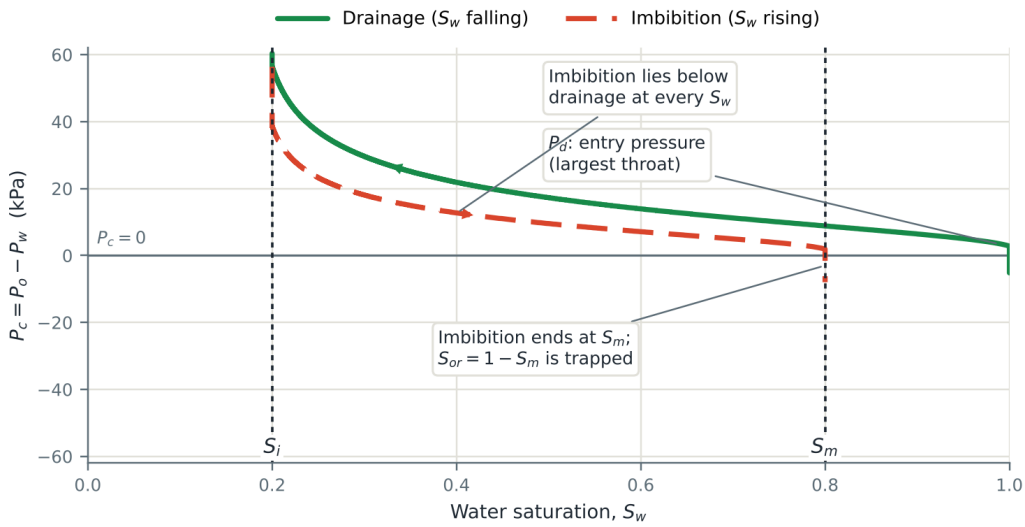


Figure 1. Drainage (solid green) and imbibition (dashed red), water-wet rock. Arrows show the direction of travel.

- **Drainage.** S_w stays at 1 until P_c reaches the entry pressure $P_d = 2\sigma\cos\theta_r / r_{max} = 2.8$ kPa (largest throat). Oil invades large pores first and small pores last, so the curve steepens and turns vertical at $S_i = 0.20$.
- **Imbibition.** Water fills the smallest pores first (highest thresholds). All thresholds are positive, so water is drawn in spontaneously. The curve lies below drainage because $\cos\theta_a < \cos\theta_r$.

- **End point.** S_m = 0.80 near P_c = 0; the other 20% of pore volume is trapped oil. Capillary forces help the waterflood.

4. Oil-wet rock

θ_r = 140°, θ_a = 160°: cos θ < 0 in every pore, so P_c is negative for the whole loop.

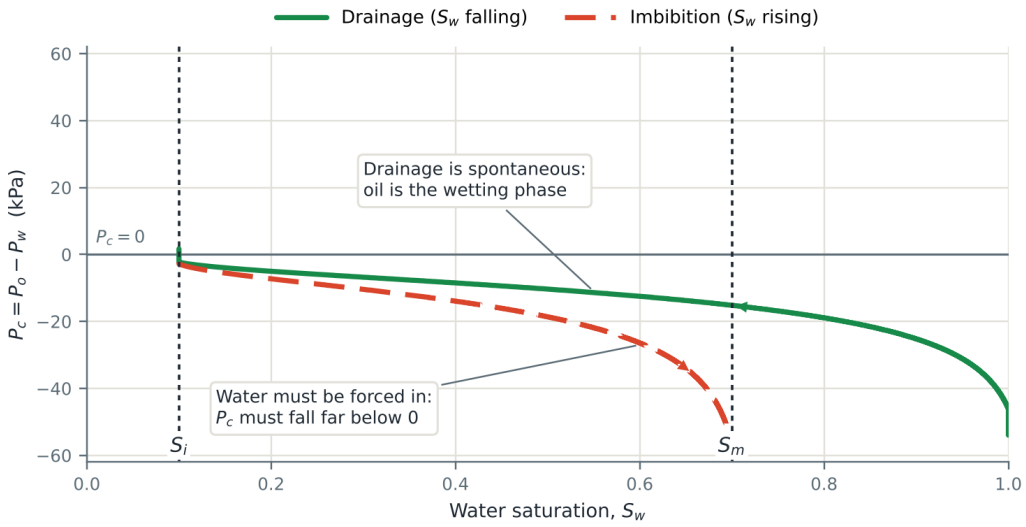


Figure 2. Drainage and imbibition, oil-wet rock. Both curves lie in the negative half.

- **Drainage.** Oil is the wetting phase and enters on its own. P_c starts at -46 kPa (smallest pore) and rises to about -2.3 kPa (largest pore) as S_w falls to S_i = 0.10. Small pores fill first, the opposite of the water-wet order.
- **Imbibition.** Water must be forced in: P_c must fall below -2.8 kPa to enter the largest pore and to -56 kPa for the smallest. Large pores fill first.
- **End point.** S_m = 0.70 (assumed S_{or} = 0.30). Capillary forces hold oil in the small pores and work against the flood.

5. Mixed-wet rock

Small pores water-wet (θ_r = 20°, θ_a = 50°), large pores oil-wet (θ_r = 130°, θ_a = 160°): P_c changes sign along each curve.

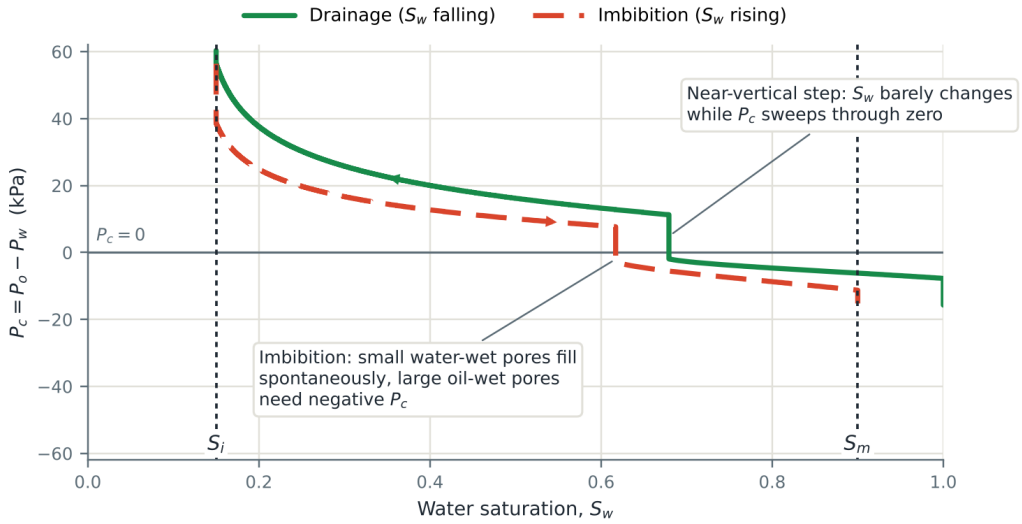


Figure 3. Drainage and imbibition, mixed-wet rock. Both curves cross P_c = 0 and show a near-vertical step.

- **Drainage.** The oil-wet large pores fill first on their own (-7.7 to −1.9 kPa). A near-vertical step follows: no pore has a threshold between −1.9 and +11 kPa, so S_w barely changes as P_c crosses zero. Above 11 kPa the water-wet small pores drain, up to 56 kPa, ending at S_i = 0.15.
- **Imbibition.** The small water-wet pores fill spontaneously (39 kPa down to about 8 kPa), then a step crosses P_c = 0, and the oil-wet large pores fill only with forced, negative P_c (−2.8 to -11.3 kPa).
- **End point.** S_m = 0.90 (assumed S_{or} = 0.10, low because oil stays connected through the large pores). Wettability indices fall between the water-wet and oil-wet extremes; real mixed-wet cores vary a lot.

6. Side-by-side comparison

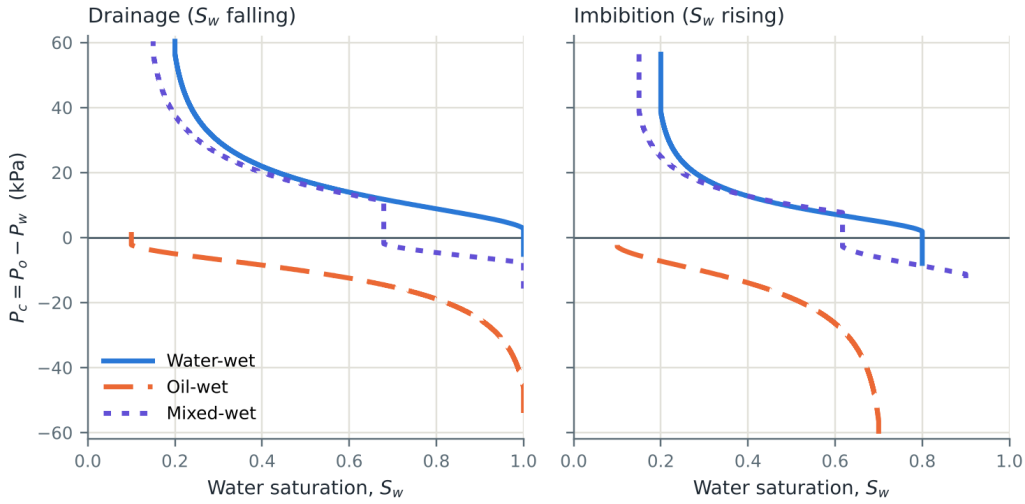


Figure 4. Drainage curves (left) and imbibition curves (right) for the three rock types.

	Water-wet	Oil-wet	Mixed-wet
Sign of P_c	positive	negative	both
Drainage	forced; large pores first	spontaneous; small pores first	large oil-wet pores spontaneously, then small pores forced
Imbibition	spontaneous; small pores first	forced; large pores first	small pores spontaneously, then large pores forced
S_i / S_m (assumed)	0.20 / 0.80	0.10 / 0.70	0.15 / 0.90

Limits

- A simple bundle of pores: no connectivity, films or throat-and-body structure. S_i, S_{or} and the contact angles are assumed, and only the main curves are shown (reversals give scanning curves inside the loop).
- Sign convention: some papers define P_c as wetting minus non-wetting pressure, which flips the signs for oil-wet rock. Check before comparing sources.
- Mercury-injection data assume a strongly water-wet surface. For oil-wet or mixed-wet reservoirs use cores with restored wettability and an Amott-Harvey or USBM test.

7. Quick answers

Why is θ_a larger than θ_r? Rough walls and chemical patches pin the contact line, so the angle depends on the direction of motion. Advancing water meets a surface that oil has contacted (adsorbed polar oil components make it less water-wet); receding water leaves a surface it has just wetted.

Lower P_o or raise P_w to imbibe? Either: only $P_o - P_w$ matters. In a waterflood the injector raises P_w; a porous-plate test holds P_w fixed and changes P_o; a centrifuge uses the density contrast.

What does the threshold mean at equilibrium? At each point on the curve, the pore at the edge of the invaded zone has a threshold equal to the applied P_c. Larger pores (lower thresholds) are already filled; smaller pores wait for a higher P_c (drainage) or a lower P_c (imbibition).